

the relay and contactor section, while some can also go to MMI usually mounted on panel doors. These also must be preferably shielded, with the shield bonded to the mounting plate at a point nearest to the PLC. For cables going to MMI, pigtails can be used for grounding, although saddle clamps are better.

Such cables should be very close to RF reference (that is, mounting plate) as these cross over to the front-panel door. This can be achieved only if the crossover happens along the bonding point of the panel to the door, that is, along the bond straps that connect the panel to the door. If possible, bond straps should be mounted along these cables or additional bond straps should be provided along these cables. Many a time, it is observed that these cables crossover along door hinges, which is not a good practice, since hinges cannot be relied upon to provide a good bond.

Motor drives are the most notorious components as far as generation of conducted and radiated emission is concerned, and are the main reasons for non-compliance observed in control panels. Most drive manufacturers give fairly-detailed information regarding the type of cables to be used for input mains supply, drive output cable going to the motor, frequency input cable and I/O and control cables. All these cables

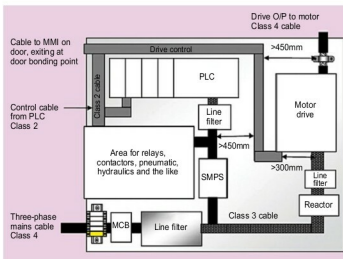


Fig. 30: Cable routing in an electronics cabinet

must be invariably shielded.

Drive manuals also discuss cable routing to be followed, how cables are installed and how saddle clamps and pigtails are to be used to ground the shields. Some even go to the extent of suggesting how much tightening torque has to be applied to the screws for proper bonding. Machine manufacturers who casually ignore these guidelines eventually face a daunting task while complying with EMC tests.

Now, output cables of motor drives can carry strongly-interfering signals and therefore classified as class 4. It is imperative that output and control cables of a drive are shielded with the shield bonded to the mounting plate by a 360-degree connector or saddle clamp at a point nearest to the drive. The shield must also be grounded at the motor end, preferably through a 360-degree bulkhead connector at the motor terminal box.

Drive control cable, which is class 2, must be separated from output cable (class 4) by a distance of more than 450mm and from mains cable (class 3) by more than 300mm. The mains cable to the drive should preferably be a shielded cable with the shield bonded to the mounting plate by a saddle clamp, and the power earth wire of the mains cable should also be fastened under this clamp. Analogue and digital signals must have their own

separate cables; there cannot be a bigger blunder than running these as wires through the same cable, even though these are of the same class (class 2).

Conclusion

Cable selection and routing is generally a subject of system-level EMC design, where a large number of components are assembled to make a system. It is generally observed that assembling all EMC-compliant components does not automatically result in a compliant system. This is because although the components that make up the system have been tested individually, their manufacturers do not know in what way their products would be assembled in a system. It is the final system assembler who has to ensure correct assembly of all components to make a compliant system. Cable selection and routing plays a vital part in this endeavor. **EFY**

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